Physics of Plasmas

Impact of the radial density profile on the Z-pinch stability at a microsecond rise time of the driving current

Cite as: Phys. Plasmas **30**, 042711 (2023); https://doi.org/10.1063/5.0132064
Submitted: 26 October 2022 • Accepted: 18 March 2023 • Published Online: 13 April 2023

R. K. Cherdizov, R. B. Baksht, V. A. Kokshenev, et al.

Impact of the radial density profile on the Z-pinch stability at a microsecond rise time of the driving current

Cite as: Phys. Plasmas **30**, 042711 (2023); doi: 10.1063/5.0132064
Submitted: 26 October 2022 · Accepted: 18 March 2023 ·
Published Online: 13 April 2023

View Online Export Citation CrossMark

R. K. Cherdizov,[1,a)] R. B. Baksht,[1] V. A. Kokshenev,[1] A. G. Rousskikh,[1] A. V. Shishlov,[1] D. L. Shmelev,[2] A. S. Zhigalin,[1] and V. I. Oreshkin[1]

AFFILIATIONS

[1]Institute of High Current Electronics of the Siberian Branch of the Russian Academy of Sciences, Tomsk 634055, Russia
[2]Institute of Electrophysics of the Ural Branch of the Russian Academy of Sciences, Ekaterinburg 620016, Russia

[a)]Author to whom correspondence should be addressed: rustam.k.cherdizov@gmail.com

ABSTRACT

This paper considers the instabilities of imploding aluminum metal-puff Z-pinches with an outer plasma shell. An experiment was performed on the GIT-12 generator (3.2–3.6 MA, $\sim$1 μs implosion times, and $\sim$15 cm initial Z-pinch radius). It was shown that the density profile of the Z-pinch material had the dominant effect on the growth and suppression of instabilities. Two Z-pinch load configurations were used. The first configuration provided a tailored density profile (TDP) [A. L. Velikovich *et al.*, Phys. Rev. Lett. **77**, 853 (1996)], which ensured the suppression of the magneto-Rayleigh–Taylor (MRT) instability in the Z-pinch. For the second configuration, the density profile was changed in such a way that a density notch from 10 to 0.5 μg/cm^3 occurred at a radius of about 3 cm from the Z-pinch axis. The notch in the density profile and the nonmonotonic increase in density resulted in a completely unstable compression of the Z-pinch. This gave rise to large-scale instabilities, which were detected by optical diagnostics. The instabilities grew and were not suppressed even in the stagnation phase, despite a sharp increase in the density of the Z-pinch material near the axis. The results were interpreted using the model proposed by Curzon *et al.* [Proc. R. Soc. London A **257**, 386 (1960)]. The total instability amplitude is the sum of the amplitudes of MRT and magnetohydrodynamic (MHD) instabilities. The growth of the total instability in the density notch region is due to the development of MRT instability. Thus, if the density profile has a notch, the Z-pinch compression in the stagnation phase occurs under strong perturbations at the magnetic field/plasma interface. This results in a dramatic growth of MHD instabilities. Hence, a stable implosion of a Z-pinch with TDP is possible only if the density increases monotonically toward the axis.

I. INTRODUCTION

The implosion dynamics of Z-pinches continues to be a subject of investigation since they are successfully used for the generation of soft x rays and in inertial fusion research.[1–4] A Z-pinch load is a cylindrical column of conductive material through which an electric current flows from a pulsed power generator (driver). The axial current creates an azimuthal magnetic field B_θ around the cylinder. The pressure of the field B_θ compresses the Z-pinch material onto the axis aligned with the current flow. During the compression, the energy stored in the Z-pinch driver is transferred to the pinch plasma, and, in an ideal case, a column of dense (up to 10^{21}–10^{22} cm^{-3}), high-temperature (up to 2–3 keV) plasma is formed on the pinch axis.[1] One of the main obstacles to achieving an ideal implosion is magneto-Rayleigh–Taylor (MRT) instabilities. According to Ref. 1, "the classical Rayleigh–Taylor (RT) instability occurs when a light fluid supports a heavy fluid against a gravitational field (g) directed from the heavy to the light fluid. In a Z-pinch, the magnetic field in the vacuum outside the pinch represents the light fluid and it accelerates the pinch inward toward the axis, so the effective gravity g points outward." An exponential increase in the amplitude of MRT instabilities in the form of spikes and bubbles sometimes leads to a complete rupture of the contracting plasma, sharply reducing the efficiency of energy transfer from the driver to the pinch plasma. In fact, it is the MRT instabilities that prevent the use of large-initial-radius Z-pinches, which are more energy-efficient.[9]

MRT instabilities can be mitigated by the snowplow mechanism.[1] This mechanism is based on the superstability of the shock wave (SW) driven into the Z-pinch load by the magnetic pressure. Velikovich *et al.*[5]

have shown that exponentially growing MRT instabilities can be completely suppressed in the case of an SW propagating into a Z-pinch load with a tailored density profile (TDP). Density tailoring affects the acceleration of the magnetic field/plasma interface.[5–7] A density profile properly tailored for given circuit parameters makes this acceleration negative[5] or zero.[6] Hammer *et al.*[6] suggested a peaked initial density profile with constant imploding sheath velocity that reduced MRT instability. A characteristic feature of this profile is the density decreasing from the pinch axis to its periphery (see Fig. 8 in Ref. 6).

According to Velikovich *et al.*,[5,11] an MRT instability will be suppressed if the radial profile of the plasma density ρ is described by the expression

$$\rho(r) = \rho_0 (r/R_0)^{-s} \tag{1}$$

with $s > 2$ and r satisfying the condition $R_0 < r < R_1$, where r is the Z-pinch radius and $R_1 \gg R_0$. As follows from expression (1), the plasma density in a pinch should increase monotonically as the radius decreases, i.e., toward the axis. It is this type of $\rho(r)$ function the plot of which is presented by Velikovich *et al.*,[5] the authors of the idea of a tailored density profile (see Fig. 1 in Ref. 5).

Sze *et al.*[7] and Levine *et al.*[8] realized the idea of a tailored density profile to suppress MRT instabilities in an imploding argon Z-pinch 12 cm in diameter for a current of 3.5 MA and an implosion time of 200 ns. The Ar triple-nozzle gas-puff Z-pinch load was designed in such a way that the initial density profile peaked at the axis.[5] As a result, in the final stage of compression (at stagnation), a 4 cm uniform plasma column was produced that generated a 7.3 ns Ar K-shell radiation pulse with a K-shell yield of 21.2 kJ. A structured Z-pinch load of a similar design was used in experiments performed on the Saturn facility.[9] For the Z-pinches with a tailored density profile produced on the Saturn facility (6.5 MA implosion current, 200 ns implosion time), stable implosion and significantly improved coupling between the Z-pinch and the generator were attained.[9] The K-shell radiation powers obtained in the experiments[7–9] are comparable to that previously obtained for an Ar Z-pinch with an implosion time of 100 ns.

Let us dwell on some details of the formation of a tailored density profile described in Refs. 7–9. In these studies, the structured Z-pinch load consisted of an outer and an inner shell imploding onto a central jet. By varying the plenum pressure in the outer and inner shells and in the central jet, a gradual increase in density toward the axis of the Z-pinch load was achieved. The mass and diameter of the central jet were chosen so that the density peaked on the Z-pinch axis. In addition, the chosen mass ratio between each shell and the central jet allowed avoiding the occurrence of a low-density region between the inner shell and the central jet. The optimal density profile [see Fig. 3(b) in Ref. 8] ensured suppression of instabilities and, as a result, a high K-radiation yield.[8] The mass accretion toward the Z-pinch axis in the process of implosion prevented the development and break-through of magnetic "bubbles" through the pinch material and, as a result, the disruption of the final plasma column. Thus, the radiating column was uniform and straight. Levine *et al.*[8] also considered different density profiles and compared the K-radiation yields obtained with them. As a result, they supposed that the high K-radiation yield was due to the absence (i.e., suppression) of instabilities. Thus, it was shown that if there was no graduate increase in mass from the periphery to the center of the Z-pinch load and, especially, if the central gas jet was absent, the K-radiation yield was significantly lower.

In recent years, several theoretical[12,13] and experimental studies[14–16,21] have been performed on the implosion dynamics of double- and triple-nozzle gas-puff Z-pinches. Extensive studies of the development of MRT instabilities in imploding triple-nozzle gas-puff Z-pinches were carried out at the COBRA facility (1.2 MA implosion current, 220 ns implosion time).[14–16,21] The authors of Refs. 14–16 were able to reduce substantially the effect of MRT instabilities on the implosion of gas-puff Z-pinches owing to the fact that the radial density profile peaked on the axis. They studied structured Z-pinch loads with an outer and an inner shell imploding onto a central jet; the experiments were carried out with noble gases. The parameters of the shells were chosen in such a way that the total mass of the pinch material would increase gradually from the periphery to the center. However, one can see (see Fig. 2 in Ref. 14), a threefold and more significant difference in the density in the regions between the shells and the density in the shells themselves (from $\sim 10^{16}$ to more than $3 \times 10^{16}\,\mathrm{cm}^{-3}$). This indicates the existence of internal low-density regions between the outer and inner shells and between the inner shell and the central jet.

The use of three-frame laser shearing interferometry and multi-frame extreme ultraviolet cameras in the experiments[14–16] made it possible to analyze in detail the occurrence and growth of MRT instabilities in Z-pinches of different gas species. In particular, for an imploded Ar Z-pinch, it was found[14] that the average instability amplitude increased from 1 mm at $t/t_{\mathrm{imp}} = 0.7$ to 2 mm at $t/t_{\mathrm{imp}} = 1$ (see Fig. 11 in Ref. 14), despite the fact that the plasma velocity in the near-axis region decreased. It can be supposed that in order to obtain stable stagnation, it is necessary to provide a monotonic increase in the function $\rho(r)$ over the entire periphery of the Z-pinch, which, perhaps, was not provided in the Z-pinch load used in the experiment.[14] If the function $\rho(r)$ oscillates, the velocity of the magnetic field/plasma interface will either increase or decrease. Therefore, it can be speculated that at some moment the magnetic field/plasma interface would turn into a state of "skin current equilibrium which was unstable to magneto-hydrodynamic (MHD) compressible modes $m = 0$ and $m = 1$" (Ref. 2). Note that already in the pioneering work by Curzon *et al.*,[17] devoted to the development of MRT instabilities in Z-pinches, it was pointed out that there is competition between the MRT and MHD modes during Z-pinch compression.

We have shown[10,18] that the TDP regime with a monotonically increasing $\rho(r)$ function provides a stable compression of a metal-puff Z-pinch with a microsecond implosion time. Here, we present the experimental results obtained for imploded metal-puff Z-pinches with both monotonically and nonmonotonically increasing $\rho(r)$ functions. These experiments have shown that the presence of an internal low-density region between the inner shell and the central jet in an imploding structured Z-pinch load with TDP leads to irreversible destruction of the final plasma column. The occurrence of an internal low-density region has the result that the plasma column is disrupted in the stagnation phase even if the plasma density sharply increases toward the pinch axis. Our results obtained in the experiment with a nonmonotonically increasing $\rho(r)$ function are in many respects in common with the results obtained earlier at the COBRA facility.[14–16] However, it should be emphasized that there was a significant difference in the methodology between our experiments and those described in

Refs. 14–16. First, in the COBRA experiments, a high-current generator with a current rise time of about 200 ns was used, whereas in our experiments, the current rise times were on the microsecond scale. It should be noted that as far as the suppression of MRT instabilities is concerned, it is preferable to use faster machines. Second, in the COBRA experiments, gas-puff Z-pinches with pre-ionization from an external source were used. In our experiments, however, the outer shell consisted of hydrocarbon plasma generated by a set of plasma guns[25] (see Sec. II). The outer shell plasma had the electron temperature $T_e \approx 2$–3 eV and the electron density $n_e \approx (1–5) \times 10^{15}\,\mathrm{cm}^{-3}$; that is, it was highly ionized. Thus, we managed to avoid the irregularities that generally occur during the breakdown of a weakly ionized gas upon application of a high voltage to a Z-pinch load. This is the obvious advantage of our methodology.

To interpret the obtained experimental results, we used the model proposed by Curzon *et al.*[17] for an equilibrium Z-pinch. According to this model, the total instability amplitude is the sum of the amplitudes of MRT and magnetohydrodynamic (MHD) instabilities. The growth of the total instability in the internal low-density region is due to the development of MRT instability. Thus, if the density profile has an internal low-density region, the Z-pinch compression in the stagnation phase occurs under strong perturbations at the magnetic field/plasma interface. This leads to a dramatic growth of MHD instabilities, just as was observed in our experiments.

The rest of the paper is arranged as follows: Sec. II describes the hardware and the diagnostics used in our experiments. The obtained experimental results are discussed in Sec. III. Section IV presents a comparison of the results obtained using the theoretical model with our experimental data, and Sec. V summarizes our conclusions.

II. EXPERIMENTAL ARRANGEMENT

The experiments were carried out on the GIT-12 generator, which consists of 12 modules. Each module is composed of a primary capacitive store, made up of nine parallel-connected Marx generators, a vacuum insulator, and a vacuum coaxial line connecting the modules to the high-voltage central collector. The vacuum line serves as an inductive divider for voltage measurements. The collector is connected to the high-voltage electrode of a low-impedance vacuum line through which the current pulse is transmitted to the Z-pinch load. For the experiment under consideration, the charge voltage of the GIT-12 generator was 50 kV and the short-circuit (SC) current was 4.7 MA with a rise time of 1.7 μs.

A characteristic feature of the load unit of the GIT-12 generator (Fig. 1) is the use of cable plasma guns in experiments with Z-pinches.[23–25] The guns, arranged in a circle of radius 17.5 cm, operate $\sim$1.8 μs before the turn-on of the GIT-12 generator and the application of high voltage to the cathode of the high-voltage (HV) gap.[25] Note that the cable guns were part of the experiment under consideration. Thus, on the operation of the GIT-12 generator in this set of experiments, a current layer was created in the Z-pinch load at a radius $R_0 = 14$–15 cm, which provided a microsecond implosion of the Z-pinch.

Figure 1 shows two configurations of the metal-puff Z-pinch load. In both cases, the metal-puff Z-pinch was produced due to a jet of Al plasma, generated by a vacuum-arc source, injected into the HV gap.[26,27] The plasma injection occurred within a time $t_{\mathrm{delay}} = 10$–12 μs before the operation of the GIT-12 generator. In the load configuration shown in Fig. 1(a), the cathode of the GIT-12 was a metal mesh with a

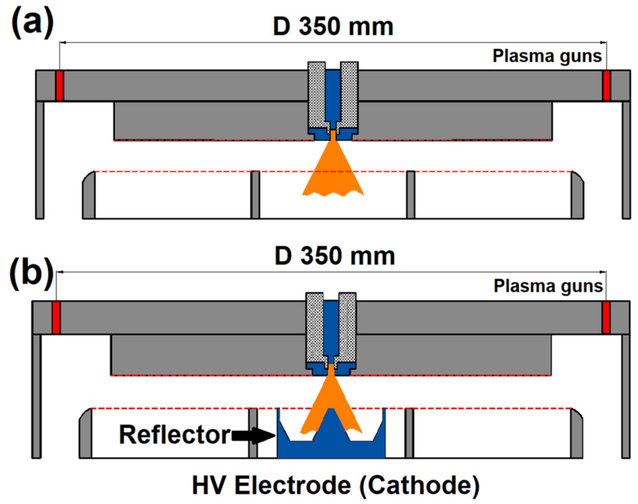

FIG. 1. Schematic diagram of the experimental Z-pinch load producing a metal-puff Z-pinch with a tailored density profile described by a monotonically (a) and a non-monotonically increasing $\rho(r)$ function (b).

transparency of 70%. According to the calculations presented in Ref. 18, the plasma propagated in the HV gap in both axial and radial directions, taking a tailored density profile described by a monotonically increasing $\rho(r)$ function (monotonic TDP). Thus, the plasma density in the pinch with a monotonic TDP increases monotonically with the decreasing radius.

In the load configuration shown in Fig. 1(b), the cathode mesh was partially replaced by a bowl-shaped reflector aligned with the HV gap. The reflector was a hollow cylinder of radius 3.3 cm with a cone-shaped protrusion on the axis. It was supposed that the plasma flow should spread along the walls of the cylinder and, due to reflection from the walls, enter the HV gap, forming a density notch at a radius of $\sim$3 cm. The reflector redistributes the material of the Al jet, so that the density increases as the radius decreases from 4 to 3.2 cm (Fig. 2).

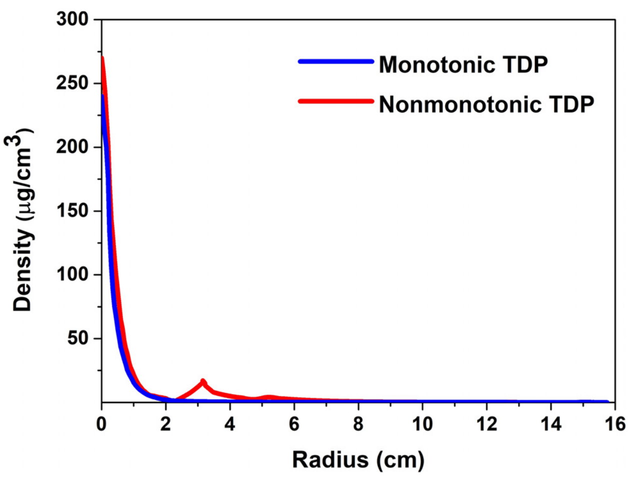

FIG. 2. Simulated density profiles for the compression of metal-puff Z-pinches in regimes with a monotonic (sh. 2778) and a nonmonotonic TDP (sh. 2781).

When the radius further decreases from 3.2 to 2.4 cm, the density decreases. Below, we will use the term "notch" for the low-density region between the radii 2.4 and 3.2 cm.

According to the simulation and experimental results presented in Refs. 18 and 24–31, the load configuration shown in Fig. 1(b) provided for the formation of a Z-pinch with a tailored density profile described by a nonmonotonically increasing $\rho(r)$ function (nonmonotonic TDP).

Figure 2 shows a monotonic and nonmonotonic initial density profiles for the compression of metal-puff Z-pinches that were plotted using the results of the simulation of an expanding plasma jet.[18] The simulation took into account multiple reflections of the plasma from the end surfaces of the cathode and anode. As a result, within the injection time t_{delay}, the plasma filled the space bounded by the electrode wire meshes and return current posts. Note that the limitation of the Z-pinch volume using wire meshes was used in previous experiments.[9,19,20] The curves given in Fig. 2 were justified by the experimental data[18] that indicated that the density described by the function $\rho(r)$ was estimated to within $\pm$20%. The experimental density profiles shown in Ref. 18 were determined using the time dependence of the pinch radius $r(t)$ constructed based on the measurements of $V(t)$ and $I(t)$.[29] With the known function $r(t)$, the density profile was determined by numerically solving the equation of motion of the Z-pinch boundary within a zero-dimensional model. Note, that such a method for determining the function $\rho(r)$ was first proposed by Hammer *et al.*[6]

As expected, in the load configuration with a reflector there was some increase in density in the range $3.2 < r < \sim 6$ cm, a density notch in the range $\sim 2.4 < r < 3.2$ cm, and an increase in density in the near-axis region.

In the experiment, we used the following diagnostics. The voltage across the Z-pinch and the current through the Z-pinch were measured (with an accuracy of 5%) using a voltage divider and inductive current grooves. The four-frame 12 bit HSFC PRO camera[22] recorded time-gated images of the pinch plasma with a frame exposure time of 3 ns. The camera captured images of the imploding and stagnated pinches in a spectral range of 300–750 nm. To detect the soft x radiation emitted by the pinch plasma, we used a filtered copper-cathode x-ray vacuum diode (XRD) ($h\nu > 1$ keV).

III. EXPERIMENTAL RESULTS

The experimental data for the monotonic and nonmonotonic TDP regimes are summarized in Table I. Table I shows the delay time of the GIT-12 generator turn-on relative to the triggering time of the source forming the jet, t_{delay}, the implosion time t_{imp}, and. the current at the implosion time, I_{imp}. The implosion time t_{imp} was measured from the point at which the current reached a level of 100 kA to the point at which the Z-pinch voltage peaked. The last column indicates the presence or absence of instabilities in the stagnation phase. We considered the plasma column to be unstable if the deviations from the average radius in the pinch image exceeded the resolution of the optical system (0.5 mm).

TABLE I. Summary of the experimental data: t_{delay} is the delay time of the GIT-12 generator turn-on relative to the triggering of the jet source, I_{imp} is the current at the implosion time, and t_{imp} is the implosion time.

Profile and shot no.	t_{delay} (μs)	I_{imp} (MA)	t_{imp} (ns)	Stability
Monotonic, 2778	12.04	3.22	935	Stable
Monotonic, 2779	12.42	3.18	907	Stable
Monotonic, 2783	12.23	3.04	884	Stable
Nonmonotonic, 2780	12.22	3.59	1139	Unstable
Nonmonotonic, 2781	10.33	3.52	1055	Unstable
Nonmonotonic, 2782	11.83	3.47	1033	Unstable

Typical waveforms of the voltage across the Z-pinch, V_{pinch}, current I_{pinch}, and XRD signal for the monotonic (sh. 2778) and nonmonotonic (sh. 2782) TDP regimes are given in Fig. 3.

We see an increase in the implosion time and, accordingly, in the Z-pinch current due to the increase in density at the periphery, for the nonmonotonic TDP regime. However, despite the increase in current, there was a significant decrease in the V_{pinch} amplitude in this regime: compare 620 kV for shot 2778 and 304 kV for shot 2782. The decrease in the V_{pinch} amplitude in itself indicates an increase in the final Z-pinch radius in the stagnation phase. Indeed, if we assume that the plasma resistance can be neglected, we have $V_{pinch} = \frac{dL_p I}{dt} = I\frac{dL_p}{dt} + L_p\frac{dI}{dt}$, where L_p is the Z-pinch inductance. Under conditions when an LC generator, such as the GIT-12 facility, serves as a driver for the pinch implosion, an increase in the pinch impedance does not entail a sharp decrease in current. The high internal inductance of the driver damps sharp current fluctuations. Indeed, as can be seen in Fig. 3(a) for shot 2778, in the final phase of compression, the current decreased by 0.11 MA within 40 ns, which corresponds to $\frac{dI}{dt} \sim 2\times\ 10^{12}$ A/s. Even for the peak value of $L_p = 15$ nH, we have $L_p\frac{dI}{dt} \sim 30$ kV, which is significantly lower than the measured V_{pinch}. Therefore, we may write

$$V_{\mathrm{pinch}} \sim I\frac{dL_p}{dt} \tag{2}$$

and

$$L_p(t) = \frac{\mu_0 l}{2\pi}\ln\frac{R_{rc}}{r_{\mathrm{ind}}}, \tag{3}$$

where R_{rc} is the radius of the return current path, μ_0 is the magnetic permeability of free space, r_{ind} is the inductive Z-pinch radius,[2,7] and l is the Z-pinch length. It follows from Eqs. (2) and (3) that a twofold difference in V_{pinch} approximately corresponds to a tenfold difference in the Z-pinch radius at the moment of implosion. In reality, in the monotonic TDP regime, the final Z-pinch radius under stable compression was five to six times less than the Z-pinch radius in the nonmonotonic TDP regime [see Figs. 4(d) and 4(h)]. In the nonmonotonic TDP regime, we also observed a prolonged stagnation phase, which is indicated by both the presence of two peaks in the $V_{pinch}(t)$ waveform and the long duration of soft x-ray emission.

Figure 4 shows successive time-gated images of the discharge plasma taken in monotonic [Figs. 4(a)–4(d)] and nonmonotonic TDP regimes [Figs. 4(e)–4(h)]. As will be shown below (Fig. 6), the shot-to-shot variability of the implosion parameters did not affect the nature and scale of the instabilities. Different intervals between the frames (20 ns for the monotonic TDP *vs* 40 ns for the nonmonotonic TDP) is explained by the fact that the implosion time for the nonmonotonic TDP was longer. Therefore, we considered it acceptable to increase the interval between the frames. It can be seen [see Figs. 4(a)–4(d)] that

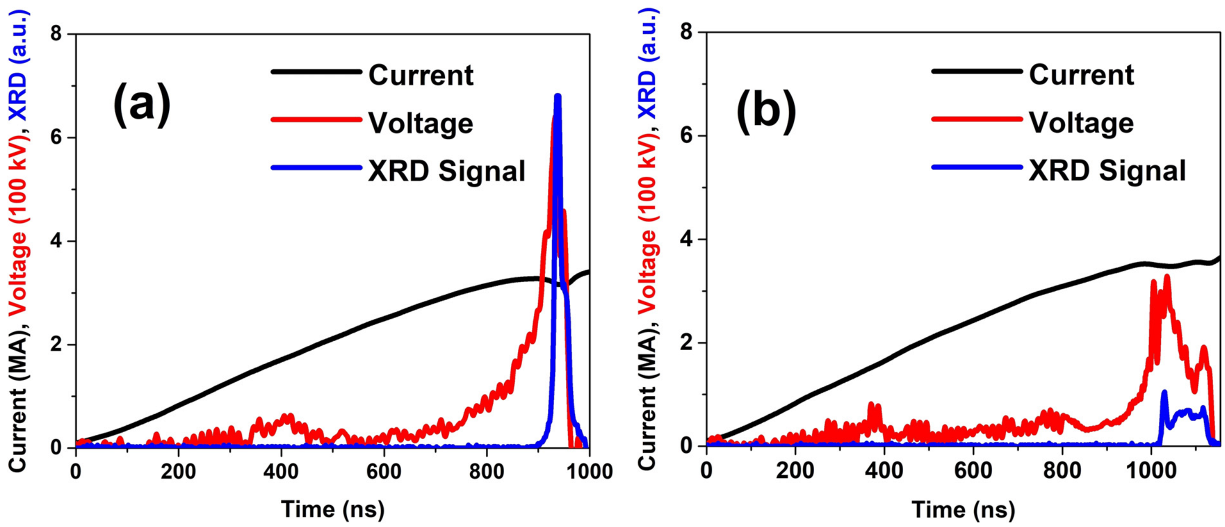

FIG. 3. Typical waveforms of the voltage across the Z-pinch, V_{pinch} (red), current I_{pinch} (black), and XRD signal (blue): (a) for the monotonic (sh. 2778) and (b) nonmonotonic (sh. 2782) TDP regimes.

the magnetic field/plasma interface was stable until the stagnation phase in the regime with a monotonic increase in density.

A completely different pattern was observed for the nonmonotonic TDP regime. The image in Fig. 4(e) (sh. 2781) was taken at $t_{frame}/t_{imp} = 0.83$, i.e., well before the stagnation phase. The image in Fig. 4(h) (sh. 2782) refers to the stagnation phase; it was taken 15 ns after maximum compression. The compression of the Z-pinch was relatively stable for 880 ns from the start of the current [see Fig. 4(e)]. This is not surprising, since the Z-pinch radius at that time was 3.2 cm and the magnetic field/plasma interface still passed through the region of a density increase, and the acceleration was negative. Note that the increase in the brightness of the luminous region in the middle of the gap in Fig. 4(e) was probably associated with the collision of the injected plasma with the plasma reflected from the reflector. However, already on the subsequent frames obtained for shot 2781 [see Figs. 4(f) and 4(g)] and shot 2782 [see Fig. 4(h)], spikes and bubbles characteristic of MRT instability are seen.

To quantitatively estimate the instability amplitude as a function of time, $\xi(t)$, we carried out the previously used procedure.[16] For each frame (that is, for a fixed point of time), points of instability maxima ("spikes") and minima (bubbles) were found along the boundary separating the bright luminous region of the Z-pinch plasma from the dark vacuum region. The points were chosen so that the tangents to them lay parallel to the Z-pinch axis. The instability amplitude $\xi(t)$ was calculated for each frame as $\sqrt{2}\overline{\sigma_r}$, where $\overline{\sigma_r}$ is the average standard deviation of the Z-pinch radius at the points of instability maxima and minima. The values of ξ for one of the unstable shots are shown in Fig. 5 (sh. 2781) together with the curves of $r_{pinch}(t) = R_{rc}\exp\left(-\frac{2\pi L_p(t)}{l\mu_0}\right)$ (Ref. 9) and the density profile $\rho(r)$. Figure 6 shows the average instability amplitude plotted based on the results of measurement processing for shots 2780, 2781, and 2782.

Figure 5 allows us to compare the increase in the instability amplitude ξ with variations in the density profile. In Fig. 5, the abscissa is the time for the function $r_{pinch}(t)$ and for the experimental data showing the instability amplitude at different points of time. The left axis is the ordinate for the function $r_{pinch}(t)$ and the right axis is the ordinate for the instability amplitude data. Figure 5 also shows the density profile $\rho(r)$ with the top axis representing the ordinate and the left axis representing the abscissa.

Before analyzing the plots presented in Fig. 5, we dwell on some features of SW propagation in the material of a pinch under compression.[32] As is known from the theory of shock waves,[36,37] ahead of the SW front, the velocity of sound is less than the SW velocity. Behind the SW front, on the contrary, the velocity of sound is greater than the SW velocity. Consequently, behind the SW front, small perturbations emanating from the magnetic piston can overtake the SW front due to the high velocity of sound, whereas ahead of the SW front (in the uncompressed material), the sound waves caused by compression do not propagate. Thus, a monotonic density profile remains monotonic ahead of the SW front.

A nonmonotonic density profile also remains unchanged ahead of the SW front. Therefore, the unimploded portion of the initial density profile $\rho(r)$ remains roughly unchanged during the implosion and corresponds to the density at the Z-pinch radius. Indeed, it is usually assumed that after the passage of an SW through a plasma, a nearly self-similar compression regime is established in the plasma column, such that the radial velocity of the material, v_r, increases linearly with radius, i.e., $v_r \propto r$ (Refs. 38 and 39). In a self-similar regime, the plasma is compressed uniformly, and the profiles of the thermodynamic functions are similar to each other at different times. Since $v_r \propto r$, the material at the plasma boundary moves faster than the material that was originally in the notch. As the pressure of the magnetic piston increases, the plasma boundary passes through the region of decreasing density. After the propagation of the shock wave, the material that was originally in the notch has a lower density but a higher temperature compared to the surrounding layers. Therefore, the pressure in the notch is the same as in the surrounding layers, and over time, both the temperature and the density are equalized due to thermal conductivity.

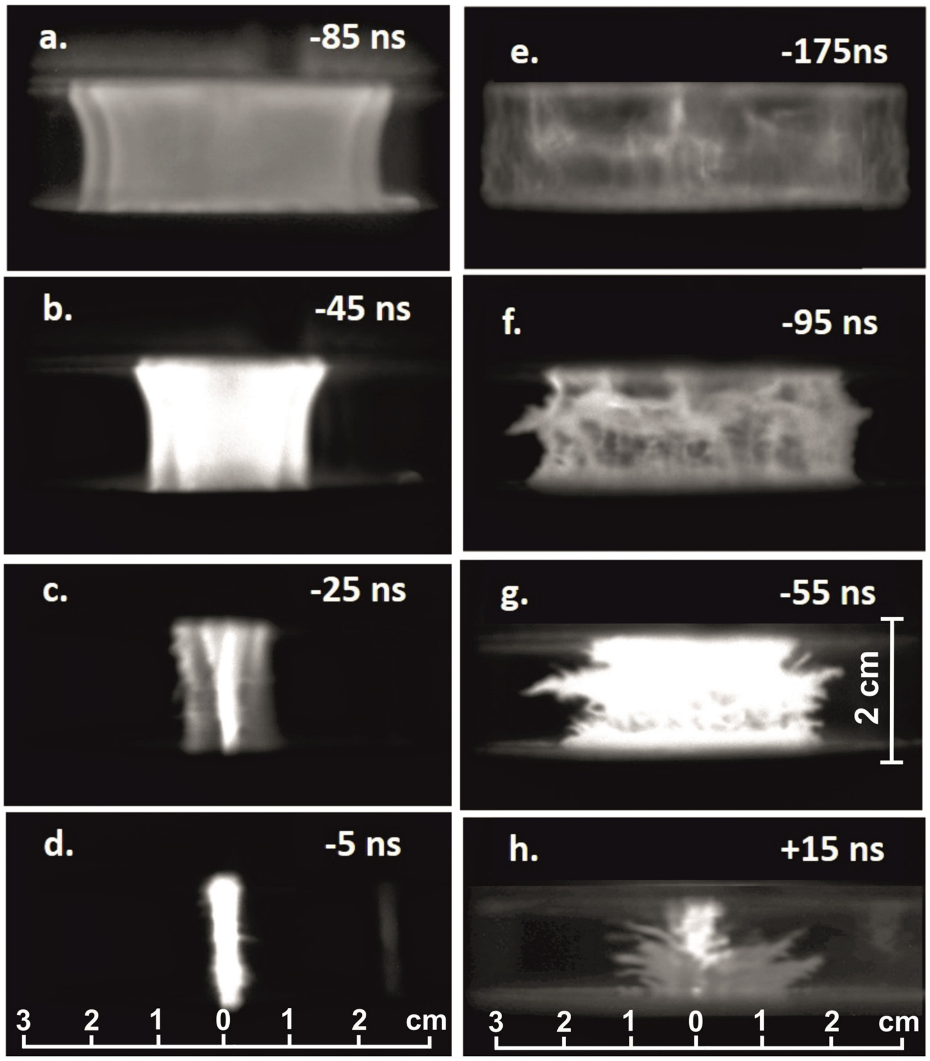

FIG. 4. Successive time-gated images of the Z-pinch plasma taken in monotonic TDP regimes: sh. 2783 (a) and (b) and sh. 2779 (c) and (d), and in nonmonotonic TDP regimes: sh. 2781 (e)–(g) and sh. 2782 (h) at the final stage of implosion. The imaging times ($t = 0$ corresponds to the implosion time, i.e., the time the Z-pinch voltage peaked) are indicated in the top right corner of each image. The implosion times for shots 2783, 2779, 2781, and 2782 were 884, 907, 1055, and 1033 ns, respectively.

As a first approximation, we can assume that the inductive radius of the plasma coincides with the averaged position of the magnetic field/plasma interface. As noted above, 880 ns after the start of the current (-175 ns with respect to t_{imp}), no instability was observed at the magnetic field/plasma interface. As the plasma was compressed, the magnetic field/plasma interface moved to a region where there was a notch in the density profile: the density decreased by almost an order of magnitude as the average radius decreased from 3.2 to 2.1–2.4 cm (see Fig. 5). Accordingly, the plasma boundary was accelerated, and, according to Ref. 11, the local acceleration directed from the plasma (heavy fluid) to the magnetic field (massless fluid) stimulated the development of MRT instabilities. Within 960 ns after the start of the current (-95 ns with respect to t_{imp}), the average instability amplitude reached 2 mm. Once the average radius became less than 2.4 cm, the density gradient $d\rho/dr$ became negative (see Fig. 5) and the instability should decay. However, even though the density increased as the Z-pinch axis was approached, the plasma boundary perturbations turned into spikes of height up to 5 mm at $t = 1000$ ns [see Figs. 4(g) and 5].

An increase in the instability amplitude with density in the near-axis region was also detected in other shots performed in nonmonotonic TDP regimes, as can be seen from the data shown in Fig. 6.

The time τ in Fig. 6 is given in relative units with respect to the average implosion time $t_{aver} = 1100$ ns for shots 2780, 2781, and 2782. According to Fig. 6, the instability amplitude increased until $\tau = 0.94t_{imp}$, which agrees with the data given in Refs. 14–16.

IV. DISCUSSION

The results of the experiment showed the following:

i. When the $\rho(r)$ distribution remained a monotonically increasing function during the compression of a Z-pinch with an initial radius of $\sim$15 cm, the Z-pinch implosion occurred stably. In this case, the final radius of the plasma column was 0.3–0.4 cm.
ii. When, for a Z-pinch with an initial radius of $\sim$15 cm, the increase in $\rho(r)$ was nonmonotonic [a density notch appeared in the $\rho(r)$ profile], the implosion became unstable. The instability

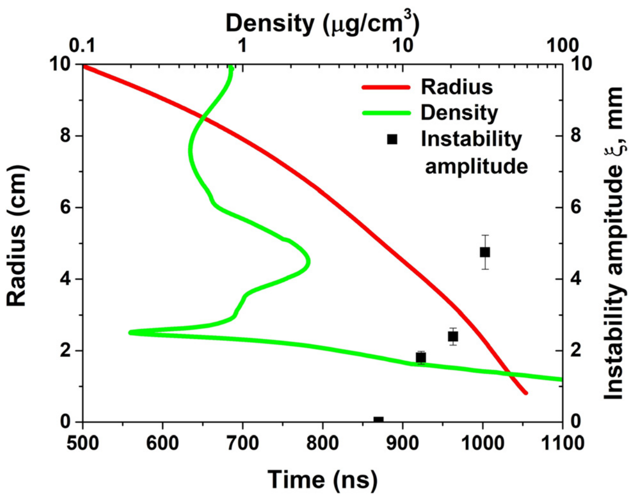

FIG. 5. Behavior of the density profile $\rho(r)$ and Z-pinch radius $r_{pinch}(t)$ for the final compression phase of a Z-pinch with a nonmonotonic TDP (sh. 2781). The implosion time was 1055 ns. The black boxes show the instability amplitude.

amplitude increased despite the sharp increase in density near the axis.

To interpret our data, we will use the model proposed by Curzon *et al.*[17] The authors of Ref. 17 suggest that there may be two independent causes for the perturbation at the magnetic field/plasma interface during the compression of a Z-pinch. One cause is the presence of a quasi-gravitational field induced by the acceleration of the plasma boundary under the action of the magnetic field, giving rise to the development of MRT instabilities. The other cause is that the balance between the plasma pressure and the magnetic field pressure at the plasma boundary is disturbed. Both causes may act simultaneously to build up identical instability modes. Some details of the model will be discussed below.

Using the r, θ, and z coordinates, we assume that a plasma column of radius r_0 is confined by a magnetic field B_θ, which is induced by a current flowing along the z axis. Perturbations will arise at the magnetic field/plasma interface. They can be described as the sum of Fourier components, each of the form $r = \hat{r}\exp\{i(kz+m\theta)+\omega t\}$ (Ref. 17), where ω and k are the angular frequency and wavenumber of the perturbation under consideration. Following Curzon *et al.*,[17] we can consider the perturbations at the plasma boundary using a simple harmonic oscillator with two independent restoring forces as an example. One of these forces causes an MRT instability; it is related to the quasi-gravity (force of inertia) that occurs when the magnetic field/plasma interface is accelerated. The second force arises in response to the perturbations that occur when the balance between the magnetic field pressure and the plasma pressure is disturbed; it causes an MHD instability. For simplicity, we assume that m = 0, thus neglecting all modes with m $\geq$ 1. For this case, the equation of motion has the form,

$$\frac{d^2 r}{dt^2} + \omega^2 r = 0, \tag{4}$$

where $\omega^2 = \omega^2_{MRT} + \omega^2_{MHD}$, and the perturbation amplitude can be determined as $\xi(t) = r(t) - r_0$, where r_0 is the equilibrium radius. According to Refs. 1 and 2, $\omega^2_{MRT} = gt$ and $\omega^2_{MHD} \propto v_A^2$, where g is the effective gravity acceleration, $g(t) = \frac{dv}{dt}$, v is the velocity of motion of the magnetic field/plasma interface, and v_A is the Alfvén velocity, which coincides in order of magnitude with the ion sound velocity v_s. Then, we can write the following relation for the instability amplitude ξ:

$$\xi(t) \propto \xi_{MRT}(t) + \xi_{MHD}(t), \tag{5}$$

where $\xi_{MRT} = \xi_0 \exp(\gamma_{MRT} t)$ is determined by the development of MRT instabilities, $\xi_{MHD} = \xi_0 \exp(\gamma_{MHD} t)$ is determined by the development of the $m = 0$ compressible mode of MHD instabilities, and the instability growth rate is determined as $\gamma = i\omega$ (Ref. 3). For MRT instabilities, the growth rate is determined by the formula

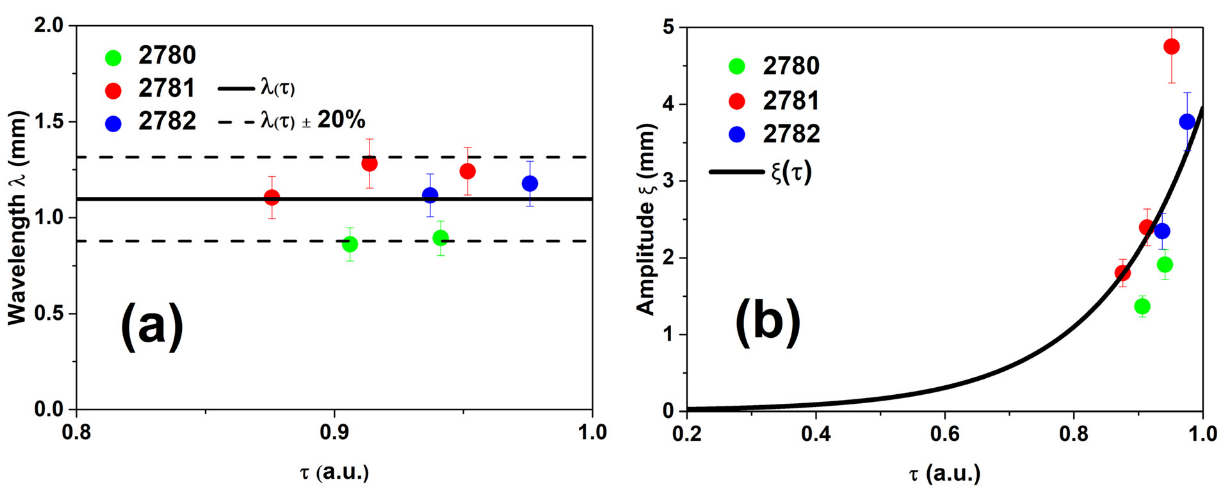

FIG. 6. The colored dots show the instability wavelength (a) and amplitude (b) along the magnetic field/plasma interface measured from the images taken by a 3 ns HSCF Pro optical camera for shots 2780, 2781, and 2782. Also, the average instability wavelength [black line, Fig. 6(a)] and average instability amplitude [black line, Fig. 6(b)] are plotted. The quantity τ is the time normalized to the average implosion time $t_{aver} = 1100$ ns. The dashed lines mark deviations of 20% from the average wavelength.

$\gamma_{\mathrm{MRT}} = \pm\sqrt{kg}$. (Ref. 2) According to Ref. 33, the pure imaginary in this formula indicates an oscillation about the magnetic field/plasma interface, i.e., stability, and the real indicates the growth of the MRT instability. Thus, we have

$$\xi_{\mathrm{MRT}} = \xi_0 \exp\left\{\left(kg\right)^{\frac{1}{2}} \times t\right\}, \tag{6}$$

where $k = \frac{2\pi}{\lambda}$ is the wavenumber of the perturbation and ξ_0 is its arbitrary (small) initial amplitude.

For the fastest growing mode of the MHD instability,[34] we have

$$\gamma_{\mathrm{MHD}} = \frac{v_s}{r_{st}}, \tag{7}$$

where r_{st} is the pinch radius at the stagnation phase and v_s is the ion sound velocity, determined as $v_s = \sqrt{\frac{2kT(1+Z)}{m_i}}$.

Figure 7 shows the functions $r_{\mathrm{ind}}(t)$, $g(t)$, and $\rho(r)$ for a monotonic TDP regime. Figure 8 shows the functions $r_{\mathrm{ind}}(t)$, $g(t)$, and $\rho(r)$ for a nonmonotonic TDP regime. The function $g(t)$ was constructed by doubly differentiating the experimentally found function $r_{\mathrm{ind}}(t)$ [see Eqs. (2) and (3)]. The dashed–dotted blue lines in Figs. 7(b) and 8(b) correspond to $g = 0$.

For the monotonic TDP regime, the value of ξ was small throughout the time of plasma compression. For this regime, even in the stagnation phase, the values of ξ did not exceed the resolution (0.5 mm) of the optical system used for taking frame-by-frame images. This is in good agreement with the plot shown in Fig. 7(b). The effective gravity $g(t)$ was negative, while its absolute value increased monotonically toward the end of the implosion, which is in good agreement with the pattern of stable implosion [see Figs. 4(a)–4(d)].

Let us consider the behavior of the instability amplitude $\xi(t)$ during pinch compression in the nonmonotonic TDP regime when the magnetic field/plasma boundary passed through the region of increasing and subsequently decreasing density. To do this, it is useful to compare the experimentally obtained values of ξ with the values of ξ_{MRT} estimated using Eq. (6). The estimates for shot 2781 are given in Table II. The first column of Table II shows the time that passed from the beginning of the current flow through the pinch, and the second column shows the values of ξ_{opt} determined using frame-by-frame imaging. The third column gives the values of ξ_{MRT} calculated as $\xi_{\mathrm{MRT}} = \xi_0 \exp\{(kg)^{\frac{1}{2}} \times t\}$ [Eq. (6)] using the experimentally obtained effective gravity $g(t)$ [see Fig. 8(b)] and the average instability wavelength $\lambda = 1.2 \times 10^{-3}$ m [see Fig. 6(a)]. Following the recommendations given in Ref. 36, we put ξ_0 equal to 0.5×10^{-4} m. Note that similar values of ξ_0 were used to estimate MRT growth rates in Ref. 15.

The data given in Table II indicate that at $t = 880$ ns both the instability amplitude obtained experimentally and that calculated using Eq. (6) are close to the optical resolution (0.5 mm). We see that at $t = 920$ and $t = 960$ ns, both ξ_{opt} and ξ_{MRT} increase significantly compared to the instability amplitude at $t = 880$ ns. It is between these time points that r_{ind} decreased from 2.6 to 2 cm, i.e., the plasma boundary passed through the region of decreasing density [see Fig. 8(a)] and the effective gravity g increased [see Fig. 8(b)]. However, at $t = 1000$ ns, ξ_{opt} still shows an increase, while ξ_{MRT} is almost zero. This can be accounted for by a change in the instability mechanism. It seems likely that MHD instabilities grew when the plasma boundary passed through the equilibrium region.

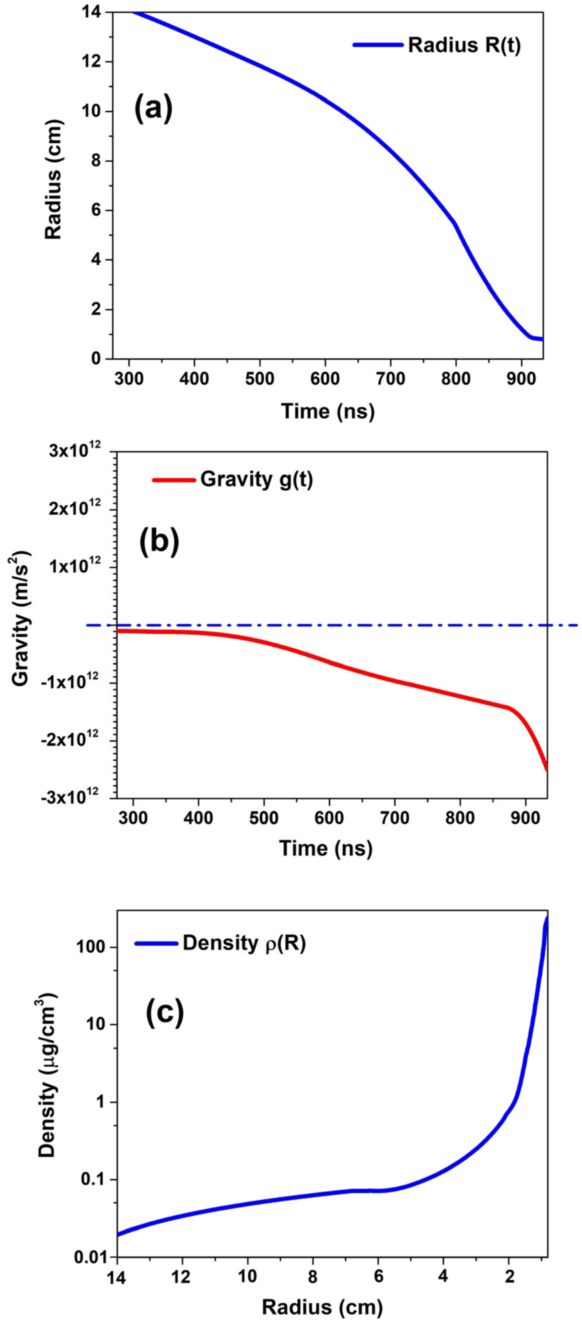

FIG. 7. Functions $r_{\mathrm{ind}}(t)$, $g(t) = \frac{dv}{dt}$, and $\rho(r)$ for a monotonic TDP (sh. 2778) The dash-dotted blue line in (b) corresponds to $g = 0$. The implosion time was 933 ns.

To estimate the growth rate of the MHD instabilities, we need to estimate the probable temperatures of the pinch in the stagnation phase. To do this, we will use the three-radial-zone model for a stagnated Z-pinch plasma.[9] According to this model, two central zones of different temperature T_e radiate in the K-shell. These zones are surrounded by a colder blanket zone. The electron temperature T_e in each

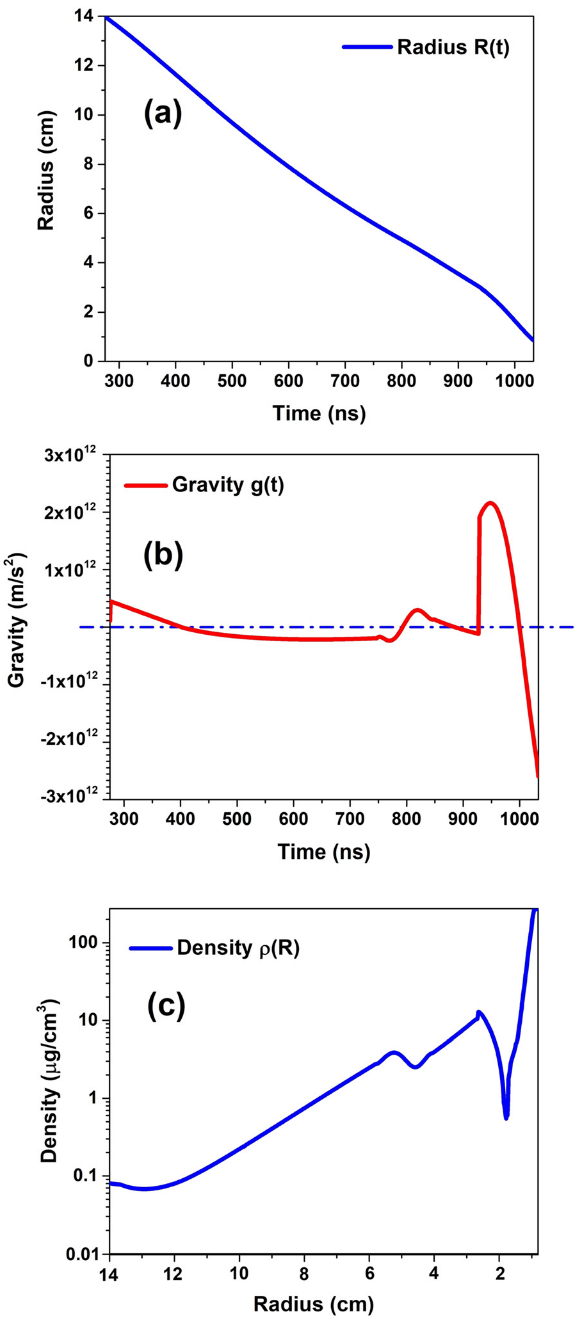

FIG. 8. Functions $r_{\text{ind}}(t)$, $g(t) = \frac{dv}{dt}$, and $\rho\,(r)$ for a nonmonotonic TDP (sh. 2782). The dash-dotted blue line in (b) corresponds to $g = 0$. The implosion time was 1033 ns.

central zone is assumed to be the temperature at which the radiation comes mostly from the K-shell range, T_{center}. For an Al plasma of density 10^{19}–$10^{20}\,\text{cm}^{-3}$, we have $T_{\text{center}} \approx 600 - 1000\,\text{eV}$ (Ref. 35). Following the model proposed in Ref. 9, we assume that the temperature T_e in the blanket zone is about half T_{center}, i.e., 300–500 eV. Thus,

TABLE II. Comparison of the instability amplitude ξ_{opt} determined using frame-by-frame imaging with ξ_{MRT} calculated as $\xi_{\text{MRT}} = \xi_0 \exp\{(kg)^{\frac{1}{2}} \times t\}$ [Eq. (6)] using experimentally obtained effective gravity $g(t)$.

Time (ns)	ξ_{opt} (mm)	ξ_{MRT} (mm)
880	~0.7	0.5
920	1.8	1.8
960	2.4	1.6
1000	4.8	0.5

according to Eq. (7), the instability growth rate at $r_{st} \approx 1$ cm is probably in the range (0.9–1.7)$\times\ 10^7\ \text{s}^{-1}$. In this case, the seeding for the growing MHD instabilities is, in fact, determined by the value of ξ_{MRT} it takes 1000 ns after the start of the current. We have $\xi_{\text{MHD}} = \xi_0 \exp\,(\gamma_{\text{MHD}} t)$, where $\xi_0 \approx 1 - 2\,\text{mm}$ and $\gamma_{\text{MHD}} = 1.7 \times 10^7\text{s}^{-1}$. Therefore, for the nonmonotonic TDP regime, 20–30 ns after the onset of stagnation, ξ_{MHD} could increase to 3–4 mm, and we see this in Table II and in Fig. 6, which shows the plot of ξ as a function of τ.

In the monotonic TDP regime, ξ_0 was so small throughout the implosion time that, in the stagnation phase, $\xi_{\text{MHD}}\,(t)$ did not increase to the optical resolution of our measurements.

V. SUMMARY

An experiment on studying the dynamics of Al metal-puff Z-pinches on the microsecond scale and the development of MRT instabilities has been performed on the GIT-12 generator. Two Z-pinch load configurations were used. In both cases, an Al metal-puff Z-pinch (R_0 ~15 cm) was produced using an Al plasma jet generated by a vacuum-arc source, which was injected into the electrode gap.

In the first load configuration, the plasma jet propagated in the axial and radial directions. Due to the multiple reflection of the plasma from the end surfaces of the cathode and anode, a tailored density profile with the material density monotonically increasing toward the axis (monotonic TDP) was formed in the Z-pinch.

In the second load configuration, part of the plasma jet material was reflected from a bowl-shaped reflector placed in line with the cathode (nonmonotonic TDP). Due to the redistribution of the Z-pinch material, a density notch was observed in the near-axis region (~3 cm in radius), but the tailored density profile persisted in the Z-pinch peripheral part. In a nonmonotonic TDP regime, a higher material density and, hence, a higher current and a longer implosion time were observed compared to a monotonic TDP regime (3.6 vs 3.2 MA and 1100 vs 920 ns, respectively).

The experiment showed that in the monotonic TDP mode, the Z-pinch compression was stable. In the nonmonotonic TDP regime, instabilities arose when the magnetic field/plasma interface passed through the internal low-density region, and they also grew in the final stage of implosion. The instabilities were not suppressed until the very end of the implosion, although the density profile peaked on the axis.

Based on the ideas put forward by Curzon *et al.*,[17] it can be supposed that the instabilities that developed in the nonmonotonic TDP regime were not suppressed due to the perturbations that arose at the magnetic field/plasma interface. These perturbations could arise from two independent causes. One cause was the presence of a quasi-gravitational field induced by the acceleration of the magnetic field/plasma interface, which led to the development of an MRT instability.

The other cause was the imbalance between the plasma pressure and the magnetic field pressure that gave rise to an MHD instability. The total instability amplitude was the sum of the MRT and MHD instability amplitudes. The growth of instabilities in the region of the density notch was associated with the growth of the MRT instability. When the density began to increase, the increase in the MRT instability amplitude stopped, but the MHD instability continued to develop. Thus, in the nonmonotonic TDP regime, the Z-pinch compression in the stagnation phase occurred under strong perturbations at the magnetic field/plasma interface. This resulted in a dramatic growth of the MHD instabilities, and the instability amplitude became comparable to the pinch radius in the stagnation phase.

Thus, to attain stable compression of a Z-pinch of large radius, the tailored density profile should provide a continuous increase in density from the outer radius to the axis.

ACKNOWLEDGMENTS

This work was supported in part by the Ministry of Science and Higher Education of the Russian Federation (State Assignment No. FWRM-2022-0001). The results discussed in Secs. III and IV were obtained with the support of the Russian Science Foundation (Grant No. 22-19-00686). The experiment was carried out using the equipment provided by the Tomsk Regional Centre for Collective Usage (Tomsk Science Center, SB, RAS). The authors are grateful to Professor A. L. Velikovich for helpful discussions and also would like to thank the referees for taking the time to read this manuscript and for many very helpful suggestions.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Rustam Cherdizov: Data curation (equal); Investigation (equal); Methodology (equal); Writing – original draft (lead); Writing – review & editing (equal). **Rina Baksht:** Conceptualization (equal); Data curation (equal); Formal analysis (lead); Writing – review & editing (lead). **V. A. Kokshenev:** Investigation (equal); Methodology (lead); Resources (lead); Writing – review & editing (equal). **Alexander Gennadievich Rousskikh:** Conceptualization (lead); Data curation (equal); Investigation (equal); Writing – review & editing (equal). **Alexander Shishlov:** Data curation (lead); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Dmitry L. Shmelev:** Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **Alexander S. Zhigalin:** Data curation (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Vladimir Ivanovich Oreshkin:** Conceptualization (equal); Formal analysis (lead); Funding acquisition (lead); Investigation (equal); Project administration (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available upon reasonable request from the authors.

REFERENCES

[1]J. L. Giuliani and R. J. Comisso, IEEE Trans. Plasma Sci. **43**, 2385 (2015).
[2]M. G. Haines, Plasma Phys. Controlled Fusion **53**, 093001 (2011).
[3]D. D. Ryutov, M. S. Derzon, and M. K. Matzen, Rev. Mod. Phys. **72**, 167 (2000).
[4]D. A. Yager-Elorriaga, M. R. Gomez, D. E. Ruiz, S. A. Slutz, A. J. Harvey-Thompson, C. A. Jennings, P. F. Knapp, P. F. Schmit, M. R. Weis, T. J. Awe, G. A. Chandler, M. Mangan, C. E. Myers, J. R. Fein, B. R. Galloway, M. Geissel, M. E. Glinsky, S. B. Hansen, E. C. Harding, D. C. Lamppa, W. E. Lewis, P. K. Rambo, G. K. Robertson, M. E. Savage, G. A. Shipley, I. C. Smith, J. Schwarz, D. J. Ampleford, K. Beckwith, K. J. Peterson, J. L. Porter, G. A. Rochau, and D. B. Sinars, Nucl. Fusion **62**, 042015 (2022).
[5]A. L. Velikovich, F. L. Cochran, and J. Davis, Phys. Rev. Lett. **77**, 853 (1996).
[6]J. H. Hammer, J. L. Eddleman, P. Springer, M. Tabak, A. Toor, K. L. Wong, G. B. Zimmerman, C. Deeney, R. Humphreys, T. J. Nash, T. W. L. Sanford, R. B. Spielman, and J. S. De Groot, Phys. Plasmas **3**, 2063 (1996).
[7]H. Sze, J. Banister, B. H. Failor, J. S. Levine, N. Qi, A. L. Velikovich, J. Davis, D. Lojewski, and P. Sincerny, Phys. Rev. Lett. **95**, 105001 (2005).
[8]J. S. Levine, J. W. Banister, B. H. Failor, N. Qi, H. M. Sze, A. L. Velikovich, R. J. Comissio, J. Davis, and D. Lojewski, Phys. Plasmas **13**, 082702 (2006).
[9]R. J. Commisso, J. P. Apruzese, D. Mosher, D. P. Murphy, B. V. Weber, J. W. Banister, B. H. Failor, J. S. Levine, N. Qi, H. M. Sze, A. Bixler, A. Bixler, P. L. Coleman, A. Jarema, J. Knight, S. Lee, M. Krishnan, J. Thompson, K. Wilson, C. A. Coverdale, and C. Deeney, in *Proceedings of the 16th IEEE International Pulsed Power Conference* (IEEE, 2007), Vol. 2, p. 1773.
[10]R. B. Baksht, V. I. Oreshkin, A. G. Rousskikh, and A. S. Zhigalin, Plasma Phys. Controlled Fusion **60**, 035015 (2018).
[11]A. L. Velikovich, F. L. Cochran, J. Davis, and Y. K. Chong, Phys. Plasmas **5**, 3377 (1998).
[12]J. Narkis, F. Conti, A. L. Velikovich, and F. N. Beg, Phys. Rev. E **104**, L023201 (2021).
[13]J. Narkis, F. Conti, and F. N. Beg, Phys. Rev. E **105**, 045205 (2022).
[14]E. S. Lavine, S. V. Rocco, J. T. Banasek, W. M. Potter, J. B. Greenly, H. Wilhelm, N. Qi, D. A. Hammer, and B. R. Kusse, Phys. Plasmas **28**, 022703 (2021).
[15]N. Qi, S. V. Rocco, J. Engelbrecht, E. S. Lavine, P. de Grouchy, J. T. Banasek, L. Atoyan, T. Byvank, W. M. Potter, J. B. Greenly, D. A. Hammer, B. R. Kusse, S. A. Pikuz, T. A. Shelkovenko, E. Kroupp, A. Fisher, and Y. Maron, IEEE Trans. Plasma Sci. **46**, 3864 (2018).
[16]P. W. L. de Grouchy, B. R. Kusse, J. Banasek, J. Engelbrecht, D. A. Hammer, N. Qi, S. Rocco, and S. N. Bland, Phys. Plasmas **25**, 072701 (2018).
[17]F. L. Curzon, A. Folkierski, R. Latham, and J. A. Nation, Proc. R. Soc. London A **257**, 386 (1960).
[18]R. K. Cherdizov, R. B. Baksht, V. A. Kokshenev, V. I. Oreshkin, A. G. Rousskikh, A. V. Shishlov, D. L. Shmelev, and A. S. Zhigalin, Plasma Phys. Controlled Fusion **64**, 015011 (2022).
[19]D. P. Murphy, R. J. Allen, B. V. Weber, R. J. Commisso, J. P. Apruzese, D. G. Phipps, and D. Mosher, Rev. Sci. Instrum. **79**, 10E306 (2008).
[20]A. Y. Labetsky, V. A. Kokshenev, N. E. Kurmaev, V. I. Oreshkin, A. G. Rousskikh, A. V. Fedyunin, F. I. Fursov, S. A. Chaikovsky, A. V. Shishlov, and N. A. Zhidkova, Plasma Phys. Rep. **34**, 228 (2008).
[21]E. S. Lavine, S. V. R. Rocco, W. M. Potter, J. Angel, E. Freeman, J. T. Banasek, J. Lawson, J. B. Greenly, H. Wilhelm, D. A. Hammer, and B. R. Kusse, Phys. Plasmas **29**, 062702 (2022).
[22]See https://spegroup.ru/upload/wikifiles/1223250.pdf for "Specifications of HSFC PRO cameras."
[23]D. Klir, S. L. Jackson, A. V. Shishlov, V. A. Kokshenev, K. Rezac, A. R. Beresnyak, R. K. Cherdizov, J. Cikhardt, B. Cikhardtova, G. N. Dudkin, J. T. Engelbrecht, F. I. Fursov, J. Krasa, J. Kravarik, P. Kubes, N. E. Kurmaev, V. Munzar, N. A. Ratakhin, K. Turek, and V. A. Varlachev, Matter Radiat. Extremes **5**, 026401 (2020).
[24]A. V. Shishlov, V. A. Kokshenev, N. E. Kurmaev, N. A. Labetskaya, F. I. Fursov, and R. K. Cherdizov, Rus. Phys. J. **62**, 1243 (2019).
[25]V. A. Kokshenev, A. Y. Labetsky, B. M. Kovalchuk, S. A. Chaikovsky, A. V. Fedunin, F. I. Fursov, N. E. Kurmaev, A. G. Rousskikh, A. V. Shishlov, and N. A. Zhidkova, in Proceedings of the 16th International Symposium on High Current Electronics (Tomsk; Publishing House, 2006), Vol. 2, p. 272.

[26]A. G. Rousskikh, A. S. Zhigalin, V. I. Oreshkin, and R. B. Baksht, Phys. Plasmas **24**, 063519 (2017).
[27]D. L. Shmelev, V. I. Oreshkin, and S. A. Chaikovsky, J. Phys. Conf. Ser. **1115**, 022014 (2018).
[28]D. L. Shmelev, A. S. Zhigalin, S. A. Chaikovsky, V. I. Oreshkin, and A. G. Rousskikh, Phys. Plasmas **27**, 092708 (2020).
[29]R. K. Cherdizov, V. A. Kokshenev, A. V. Shishlov, R. B. Baksht, V. I. Oreshkin, A. G. Rousskikh, and A. S. Zhigalin, in *Proceedings of the 7th International Congress on Energy Fluxes and Radiation Effects (EFRE)* (IEEE, 2020), p. 55.
[30]A. G. Rousskikh, A. V. Fedyunin, A. P. Artyomov, A. S. Zhigalin, and V. I. Oreshkin, Curr. Appl. Phys. **19**, 704 (2019).
[31]V. I. Oreshkin, R. B. Baksht, R. K. Cherdizov, E. V. Oreshkin, N. A. Ratakhin, A. G. Rousskikh, A. V. Shishlov, V. A. Vankevich, and A. S. Zhigalin, Plasma Phys. Controlled Fusion **63**, 045022 (2021).
[32]J. R. Angus, A. J. Link, and A. E. W. Schmidt, Phys. Plasmas **27**, 012108 (2020).
[33]Y. Zhou, R. J. Williams, P. Ramaprabhu, M. Groomd, B. Thornber, A. Hillier, W. Mostert, B. Rollin, S. Balachandar, P. D. Powell, A. Mahalov, and N. Attal, Physica D **423**, 132838 (2021).
[34]S. Ishimaru, *Basics Principles of Plasma Physics* (CRC Press, 1973).
[35]C. A. Coverdale, B. Jones, D. J. Ampleford, J. Chittenden, C. Jennings, J. W. Thornhill, J. P. Apruzese, R. W. Clark, K. G. Whitney, A. Dasgupta, J. Davis, J. Guiliani, P. D. LePell, C. Deeney, D. B. Sinars, and M. E. Cuneo, High Energy Density Phys. **6**, 143 (2010).
[36]A. L. Velikovich, Phys. Fluids B **3**, 492 (1991).
[37]Y. B. Zel'dovich and Y. P. Raizer, *Physics of Shock Waves and High-Temperature Hydrodynamic Phenomena* (Courier Corporation, 2002).
[38]M. A. Liberman and A. L. Velikovich, Nucl. Fusion **26**, 709 (1986).
[39]F. S. Felber, Phys. Fluids **25**, 643 (1982).